Machine Learning 1
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Due 2/28/26

Assignment 3

In the Week 6 lecture, we introduced the supervised binning method. We trained a CART decision tree of depth 2 for each predictor and used the resulting bins as categorical predictors in a multinomial logistic regression model. That model predicts Dropout with 74.5% accuracy, Enrolled with 21.3%, and Graduate with 88.6%.

In Week 7, you explored the CHAID algorithm for training decision trees. You'll have the opportunity to apply this algorithm to supervised binning and then use the resulting bins as categorical predictors to build a multinomial logistic regression model.

You use the Student Academic Success.csv, which was used in the lectures. The categorical predictors are **Attendance**, **Course**, **Debtor**, **Displaced**, **Educational special needs**, **Gender**, **International**, **Marital status**, **Scholarship holder**, and **Tuition fees up to date**. The continuous predictors are **Age at enrollment** and **Curricular units 1st sem (approved)**. The label variable is **Target**, with three categories: *Dropout*, *Enrolled*, and *Graduate*. The reference category is *Graduate*.

Question 1: You'll build a simple CHAID tree with a depth of 1 for each categorical predictor that has three or more categories. Afterward, you'll create a table showing the counts of terminal nodes for each of these trees. In this table, each row will represent a node, and you'll include the original predictor categories for each node to make everything clear. Each column will correspond to a label category, helping you easily see the distribution.

Node ID	Course Codes (Groups)	Dropout	Enrolled	Graduate	Total N	Majority Outcome
1	1, 7	100.0	67.0	15.0	182.0	**Dropout**
2	2, 4, 11, 13, 14	392.0	180.0	459.0	1,031.0	**Graduate**
3	3, 15	172.0	55.0	319.0	546.0	**Graduate**
4	5, 6	141.0	117.0	305.0	563.0	**Graduate**
5	8, 17	214.0	75.0	120.0	409.0	**Dropout**
6	9, 16	219.0	158.0	195.0	572.0	**Dropout**
7	10, 12	183.0	142.0	796.0	1,121.0	**Graduate**
Root	**All Courses**	**1,421**	**794**	**2,209**	**4,424**	**Graduate**

Node ID	Marital Status Groups	Dropout	Enrolled	Graduate	Total N	Dropout Rate
1	1, 3	1,185.0	722.0	2,016.0	3,923.0	**30.2%**
2	2, 4, 5, 6	236.0	72.0	193.0	501.0	**47.1%**
Root	**Global Total**	**1,421.0**	**794.0**	**2,209.0**	**4,424.0**	**32.1%**

Question 2: You'll train a CHAID tree with a depth of 1 for each continuous predictor, considering each unique value as its own category. To make sure each bin has enough observations, you'll set a maximum limit for each predictor. For instance, for **Age at enrollment**, values over 55 will be capped at 55 (meaning whichever is smaller between 55 and the actual **Age at enrollment**). Similarly, for **Curricular units 1st sem (approved)**, the cap will be at 15.

Afterward, you'll create a table showing the counts of terminal nodes for each of these trees. In this table, each row will represent a node, and you'll include the original predictor values for each node to make everything clear. Each column corresponds to a label category, making it easy to see the distribution.

Node ID	Age Group	Dropout	Enrolled	Graduate	Total N	Dropout %	Graduate %
1	17–19	409.0	331.0	1,212.0	1,952.0	20.9%	**62.1%**
2	20	133.0	140.0	326.0	599.0	22.2%	**54.4%**
3	21–22	151.0	107.0	238.0	496.0	30.4%	**48.0%**
4	23–24	97.0	54.0	88.0	239.0	40.6%	36.8%
5	25–28	201.0	67.0	93.0	361.0	55.7%	25.8%
6	29–31	111.0	18.0	41.0	170.0	**65.3%**	24.1%
7	32–55	319.0	77.0	211.0	607.0	52.6%	34.8%

Node ID	Unit Bins (Credits/Courses)	Dropout	Enrolled	Graduate	Total N	Grad %
1	0	570.0	71.0	77.0	718.0	10.7%
2	1–2	205.0	75.0	7.0	287.0	2.4%
3	3	139.0	93.0	37.0	269.0	13.8%
4	4	158.0	145.0	130.0	433.0	30.0%
5	5	145.0	193.0	385.0	723.0	53.3%
6	6–8	160.0	194.0	1,396.0	1,750.0	**79.8%**
7	9–12	27.0	19.0	111.0	157.0	**70.7%**
8	13–14	15.0	3.0	22.0	40.0	55.0%
9	15	2.0	1.0	44.0	47.0	**93.6%**

Question 3: You will train a multinomial logistic regression using the Forward Selection method with an entry threshold of 5%. The predictors are the original categorical predictors with exactly two categories, as well as the CHAID tree leaves. You should create a step summary table that includes step numbers. At each step, include the predictor's name, the log-likelihood value, the model degrees of freedom, the Chi-square test statistic, the test degrees of freedom, and the significance value of the test.

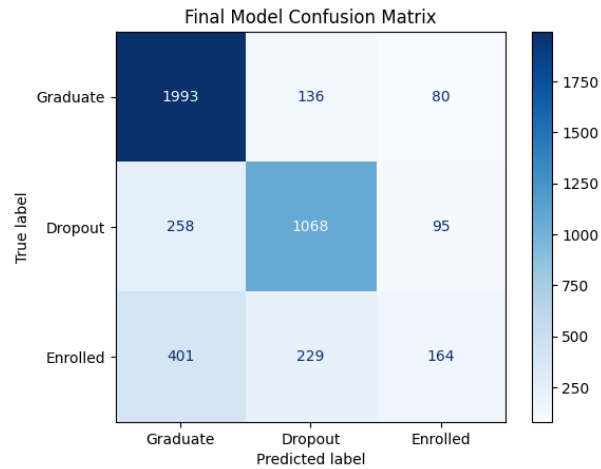
	Step	Predictor	Log-Likelihood	Model DF	Chi-square	Test DF	Significance
0	1	Curricular units group	-3431.956909	18	2159.751759	16	0.000000e+00
1	2	Tuition fees up to date	-3198.486928	20	466.939963	2	4.029725e-102
2	3	Age group	-3107.181567	32	182.610722	12	1.241823e-32
3	4	Scholarship holder	-3054.904669	34	104.553794	2	1.978937e-23
4	5	Course group	-3009.826178	46	90.156983	12	4.603661e-14
5	6	Debtor	-2994.123663	48	31.405030	2	1.515250e-07
6	7	Daytime/evening attendance	-2980.909844	50	26.427637	2	1.825204e-06
7	8	Gender	-2974.163569	52	13.492551	2	1.175249e-03
8	9	International	-2970.935556	54	6.456026	2	3.963617e-02

Question 4: You will present the confusion matrix and the row accuracy for the Question 3 model. How does this model compare to the one discussed in the Week 6 lecture?

Confusion Matrix with Row Accuracy:

Predicted Actual	Dropout	Enrolled	Graduate	Row Accuracy (%)
Dropout	**1068**	**95**	**258**	**75.16**
Enrolled	**229**	**164**	**401**	**20.65**
Graduate	**136**	**80**	**1993**	**90.22**

Overall Model Accuracy: 72.90%



This model performs slightly better at predicting dropout and graduate, but slightly worse at predicting enrolled. Additionally, the forward selection model only uses significant predictors instead of every predictor making it more efficient.

Category	Week 6 Model (CART Binning)	Forward Selection Model	Difference
Dropout	74.5%	**75.16%**	**+0.66%**
Enrolled	**21.3%**	20.65%	-0.65%
Graduate	88.6%	**90.22%**	**+1.62%**

Question 5: Building on your model from Question 3, could you find out which predictor value combinations would lead the model to predict the highest probability for each label category?

	Target Category	Graduate	Dropout	Enrolled	Curricular units group	Tuition fees up to date	Age group	Scholarship holder	Course group	Debtor	Daytime/evening attendance	Gender	International
1550	Graduate	0.992395	0.004708	0.002897	8.0	1.0	6.0	1.0	2.0	0.0	0.0	0.0	0.0
842	Dropout	0.000344	0.996630	0.003026	0.0	0.0	5.0	0.0	4.0	1.0	1.0	1.0	0.0
2515	Enrolled	0.098361	0.141025	0.760614	3.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	1.0